Practical - 4

Figures and Graphics

Customising Fonts and Colours

1 LaTeX Code

The following is the complete LaTeX source code used to create this practical document.

```
\documentclass[12pt,a4paper]{article}

% Packages
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{lmodern}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{graphicx}
\usepackage{float}
\usepackage{wrapfig}
\usepackage{xcolor}
\usepackage{listings}
\usepackage{fancyhdr}
\usepackage{caption}

% Colours
\definecolor{navy}{RGB}{25,55,100}
\definecolor{teal}{RGB}{0,125,130}
\definecolor{lightblue}{RGB}{230,240,250}
\definecolor{codegreen}{rgb}{0,0.5,0}
\definecolor{codegray}{rgb}{0.5,0.5,0.5}
\definecolor{backcolour}{rgb}{0.95,0.95,0.95}

% Code listing style
\lstdefinestyle{latexstyle}{
    backgroundcolor=\color{backcolour},
    commentstyle=\color{codegreen},
    keywordstyle=\color{blue},
    numberstyle=\tiny\color{codegray},
    basicstyle=\ttfamily\scriptsize,
    breaklines=true,
    keepspaces=true,
    numbers=left,
    numbersep=8pt,
    showstringspaces=false,
    frame=single,
    tabsize=4
}

\lstset{style=latexstyle}

% Header and footer
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{Figures, Fonts and Colours}
\fancyhead[R]{Practical - 4}

\fancyfoot[R]{Page \thepage}
\renewcommand{\headrulewidth}{0.4pt}
\renewcommand{\footrulewidth}{0.4pt}

\setlength{\parindent}{0pt}
\setlength{\parskip}{6pt}

\captionsetup{font=small,labelfont=bf}

\begin{document}

% Cover Page
\begin{titlepage}
    \pagecolor{lightblue}
    \centering
    \vspace*{2cm}

    \vspace{2cm}

    {\Huge\bfseries\color{navy}
```

```
    Practical - 4\par}

    \vspace{1cm}

    {\LARGE\bfseries Figures and Graphics\par}

    \vspace{0.5cm}

    {\Large\color{teal}
    Customising Fonts and Colours\par}

    \vfill

    \vspace{1cm}
    \nopagecolor
\end{titlepage}

% LaTeX Source Code
\section{LaTeX Code}

The following is the complete LaTeX source code used
to create this practical document.

\lstinputlisting[
    style=latexstyle,
    language={[LaTeX]TeX}
]{prac_4.tex}

\clearpage

% Practical Content
\section{Figures and Graphics}

Figures are used to present information visually.
LaTeX allows images to be inserted, resized, positioned
and displayed with captions.

\subsection{Inserting an Image}

The \texttt{graphicx} package provides the
\verb|\includegraphics| command to insert images.

Example:

\begin{verbatim}
\begin{figure}[H]
    \centering
    \includegraphics[width=0.6\textwidth]{image.png}
    \caption{Example Image}
\end{figure}
\end{verbatim}

Replace \texttt{image.png} with the filename of your
own image.

\subsection{Resizing and Positioning}

The width of an image can be changed using
\texttt{width}, while \verb|\centering| places it
in the centre of the page.

\begin{figure}[H]
    \centering
    \fcolorbox{teal}{lightblue}{
        \begin{minipage}[c][3cm][c]{0.6\textwidth}
            \centering
            {\Large\bfseries\color{navy}
            Sample Figure}
        \end{minipage}
    }
    \caption{A sample figure with a caption}
    \label{fig:sample}
\end{figure}
```

```
142
143 Figure \ref{fig:sample} demonstrates figure positioning
144 and caption formatting.
145
146 \subsection{Wrapping Text Around an Image}
147
148 The \texttt{wrapfig} package allows text to flow beside
149 an image.
150
151 Example:
152
153 \begin{verbatim}
154 \begin{wrapfigure}{r}{0.35\textwidth}
155     \centering
156     \includegraphics[width=0.3\textwidth]{image.png}
157     \caption{Small Image}
158 \end{wrapfigure}
159 \end{verbatim}
160
161 The letter \texttt{r} places the image on the right.
162 Use \texttt{l} to place it on the left.
163
164 \section{Customising Fonts and Colours}
165
166 \subsection{Font Styles}
167
168 LaTeX provides commands to change font styles.
169
170 \begin{itemize}
171     \item \textbf{Bold text}
172     \item \textit{Italic text}
173     \item \underline{Underlined text}
174     \item \textsf{Sans-serif font}
175     \item \texttt{Monospaced font}
176 \end{itemize}
177
178 \subsection{Font Sizes}
179
180 Different font sizes can be selected using commands.
181
182 {\small Small text}
183
184 {\normalsize Normal text}
185
186 {\large Large text}
187
188 {\Large Larger text}
189
190 \subsection{Changing Colours}
191
192 The \texttt{xcolor} package is used to change text
193 and background colours.
194
195 Examples:
196
197 \textcolor{navy}{This text is navy blue.}
198
199 \textcolor{teal}{This text is teal.}
200
201 \colorbox{lightblue}{This text has a coloured background.}
202
203 Custom colours can be defined using RGB values:
204
205 \begin{verbatim}
206 \definecolor{myblue}{RGB}{25,75,130}
207 \textcolor{myblue}{Coloured text}
208 \end{verbatim}
209
210 \subsection{Custom Cover Page}
211
212 A cover page can be customised using different fonts,
213 colours and images.
214
```

```
Commands such as \verb|\Huge|, \verb|\bfseries|,
\verb|\color| and \verb|\includegraphics| can be used
to change text size, style, colour and insert images.

The cover page of this practical demonstrates the use
of custom fonts, colours and spacing.

\section{Conclusion}

In this practical, we learned how to insert figures,
resize and position images, add captions and wrap text
around images. We also learned how to customise font
styles, font sizes and colours and create a custom
cover page using LaTeX.

\end{document}
```

2 Figures and Graphics

Figures are used to present information visually. LaTeX allows images to be inserted, resized, positioned and displayed with captions.

2.1 Inserting an Image

The `graphicx` package provides the `\includegraphics` command to insert images.

Example:

```
\begin{figure}[H]
    \centering
    \includegraphics[width=0.6\textwidth]{image.png}
    \caption{Example Image}
\end{figure}
```

Replace `image.png` with the filename of your own image.

2.2 Resizing and Positioning

The width of an image can be changed using `width`, while `\centering` places it in the centre of the page.

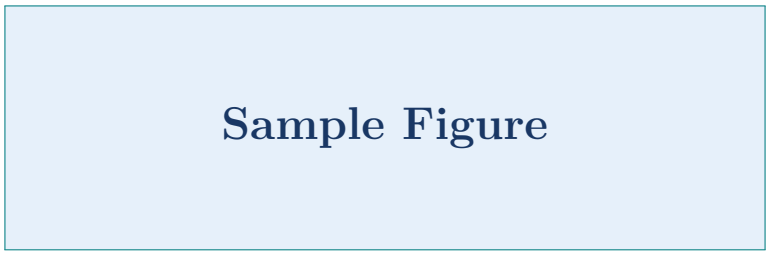


Figure 1: A sample figure with a caption

Figure 1 demonstrates figure positioning and caption formatting.

2.3 Wrapping Text Around an Image

The `wrapfig` package allows text to flow beside an image.

Example:

```
\begin{wrapfigure}{r}{0.35\textwidth}
    \centering
    \includegraphics[width=0.3\textwidth]{image.png}
    \caption{Small Image}
\end{wrapfigure}
```

The letter `r` places the image on the right. Use `l` to place it on the left.

3 Customising Fonts and Colours

3.1 Font Styles

LaTeX provides commands to change font styles.

- **Bold text**
- *Italic text*
- Underlined text
- Sans-serif font
- `Monospaced font`

3.2 Font Sizes

Different font sizes can be selected using commands.

Small text

Normal text

Large text

Larger text

3.3 Changing Colours

The `xcolor` package is used to change text and background colours.

Examples:

This text is navy blue.

This text is teal.

This text has a coloured background.

Custom colours can be defined using RGB values:

```
\definecolor{myblue}{RGB}{25,75,130}
\textcolor{myblue}{Coloured text}
```

3.4 Custom Cover Page

A cover page can be customised using different fonts, colours and images.

Commands such as `\Huge`, `\bfseries`, `\color` and `\includegraphics` can be used to change text size, style, colour and insert images.

The cover page of this practical demonstrates the use of custom fonts, colours and spacing.

4 Conclusion

In this practical, we learned how to insert figures, resize and position images, add captions and wrap text around images. We also learned how to customise font styles, font sizes and colours and create a custom cover page using LaTeX.